POWER

File: POWER.kicad_sch

Driver

File: Driver.kicad_sch

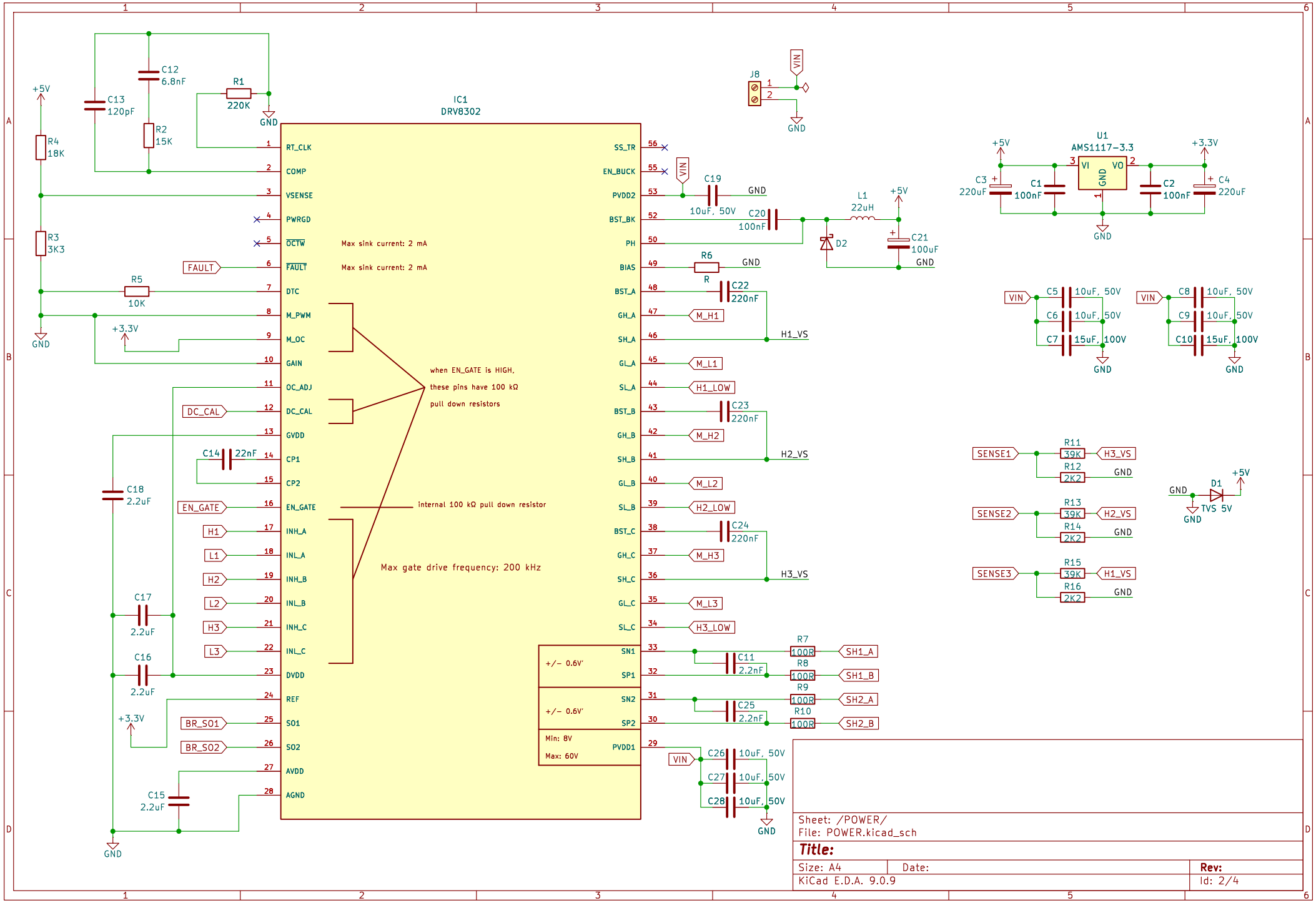
MOSFETS

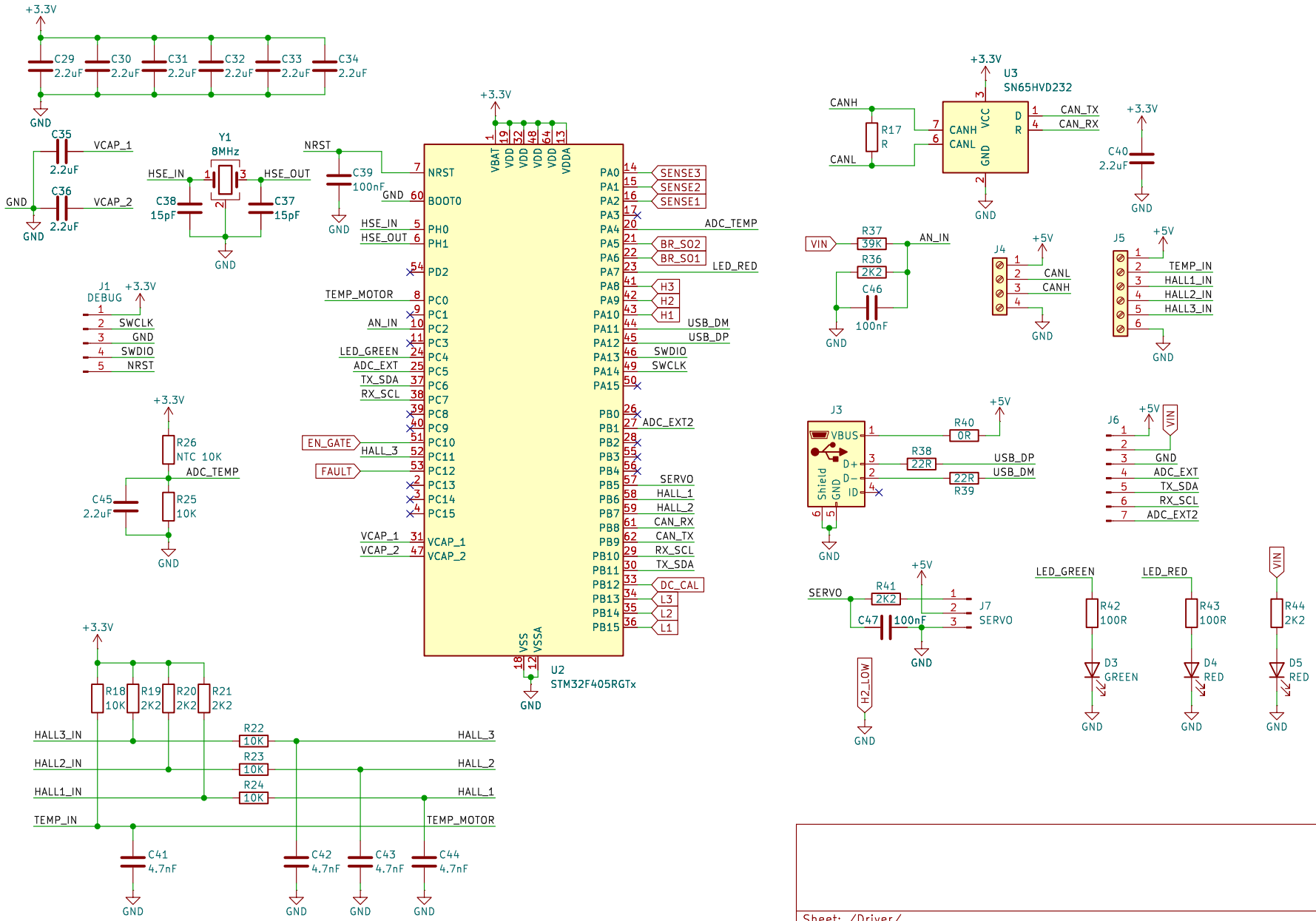
File: MOSFETS.kicad_sch

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File: ESC.kicad_sch

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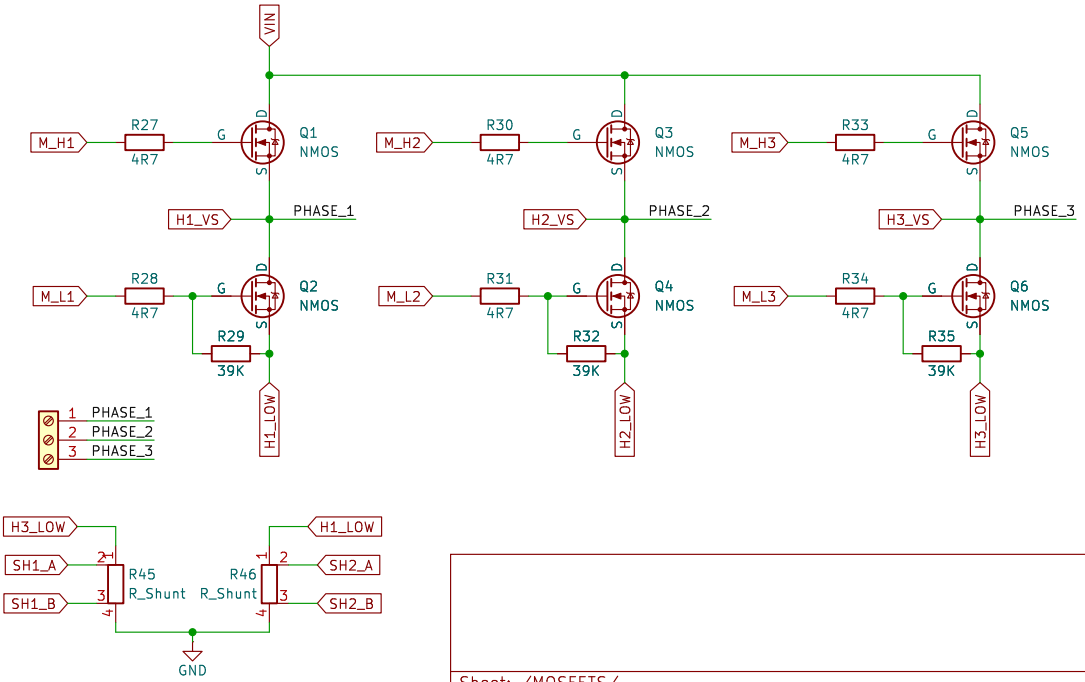

IC1
DRV8302
RT_CLK
COMP
VSENSE
PWRGD
OCTW
FAULT
DTC
M_PWM
M_OC
GAIN
OC_ADJ
DC_CAL
GVDD
CP1
CP2
EN_GATE
INH_A
INL_A
INH_B
INL_B
INH_C
INL_C
DVDD
REF
SO1
SO2
AVDD
AGND
SS_TR
EN_BUCK
PVDD2
BST_BK
PH
BIAS
BST_A
GH_A
SH_A
GL_A
SL_A
BST_B
GH_B
SH_B
GL_B
SL_B
BST_C
GH_C
SH_C
GL_C
SL_C
SN1
SP1
SN2
SP2
PVDD1
Max sink current: 2 mA
Max sink current: 2 mA
when EN_GATE is HIGH,
these pins have 100 kΩ
pull down resistors
internal 100 kΩ pull down resistor
Max gate drive frequency: 200 kHz
+/- 0.6V
+/- 0.6V
Min: 8V
Max: 60V
U1
AMS1117-3.3
VI
VO
GND
J8
VIN
+5V
+3.3V
GND
C1 100nF
C2 100nF
C3 220uF
C4 220uF
C5 10uF, 50V
C6 10uF, 50V
C7 15uF, 100V
C8 10uF, 50V
C9 10uF, 50V
C10 15uF, 100V
C11 2.2nF
C12 6.8nF
C13 120pF
C14 22nF
C15 2.2uF
C16 2.2uF
C17 2.2uF
C18 2.2uF
C19 10uF, 50V
C20 100nF
C21 100uF
C22 220nF
C23 220nF
C24 220nF
C25 2.2nF
C26 10uF, 50V
C27 10uF, 50V
C28 10uF, 50V
R1 220K
R2 15K
R3 3K3
R4 18K
R5 10K
R6 R
R7 100R
R8 100R
R9 100R
R10 100R
R11 39K
R12 2K2
R13 39K
R14 2K2
R15 39K
R16 2K2
L1 22uH
D1 TVS 5V
D2
FAULT
DC_CAL
EN_GATE
H1
L1
H2
L2
H3
L3
BR_SO1
BR_SO2
M_H1
M_L1
H1_LOW
M_H2
M_L2
H2_LOW
M_H3
M_L3
H3_LOW
H1_VS
H2_VS
H3_VS
SH1_A
SH1_B
SH2_A
SH2_B
SENSE1
SENSE2
SENSE3
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Title:

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